

Training Data Audit

Verifying `us_shocks.csv` and `us_labels.csv` for Codebook Development

Esteban Degetau
World Bank
estebandegetau@gmail.com

Agustín Samano
World Bank
asamanope-
naloza@worldbank.org

May 30, 2026

Overview

This notebook audits the two ground-truth datasets that anchor Phase 0 codebook development. Both files were re-entered from the Romer & Romer (2010) companion paper using an automated parser ([python/parse_rr_companion.py](#)) and verified against the source text for 12 spot-checked acts with zero discrepancies.

The companion paper documents **49 act headings** covering the postwar era (1945–2003). Our `us_shocks.csv` uses the **standard series** (excluding retroactive changes), which is the baseline R&R recommend for estimating the macroeconomic effects of tax changes.

1. Act Counts

Dataset Dimensions

Dataset	Unique Acts	Total Rows
<code>us_shocks</code>	49	90
<code>us_labels</code>	40	376

The shock data contains **49 acts across 90 rows** because many acts have multiple quarterly entries (phased-in tax changes, different provisions taking effect at different times). The labels dataset contains **372 evidence passages across 40 acts**. Nine acts lack extracted evidence passages; these are mostly Social Security amendments and highway acts whose companion paper narratives contain few direct quotations from policymakers.

2. Acts in us_shocks

All Acts in us_shocks.csv

Act Name	Date Signed	Quarters	Motivation(s)	Exogeneity
Revenue Act of 1945	1945-11-08	1	Spending-driven	Endogenous
Social Security Amendments of 1947	1947-08-06	1	Deficit-driven	Exogenous
Revenue Act of 1948	1948-04-02	1	Long-run	Exogenous
Social Security Amendments of 1950	1950-08-28	2	Spending-driven, Deficit-driven	Endogenous, Exogenous
Revenue Act of 1950	1950-09-23	1	Spending-driven	Endogenous
Excess Profits Tax Act of 1950	1951-01-03	1	Spending-driven	Endogenous
Revenue Act of 1951	1951-10-20	1	Spending-driven	Endogenous
Expiration of Excess Profits Tax and of Tem-	1954-01-01	1	Spending-driven, Deficit-driven	Endogenous, Exogenous

All Acts in us_shocks.csv

Act Name	Date Signed	Quarters	Motiva- tion(s)	Exogene- ity
porary Income Tax In- creases Excise Tax Re- duction Act of 1954	1954-04-01	1	Spend- ing-dri- ven	Endoge- nous
Internal Revenue Code of 1954	1954-08-16	1	Long-run	Exoge- nous
Social Se- curity Amend- ments of 1954	1954-09-01	1	Spend- ing-dri- ven	Endoge- nous
Federal- Aid High- way Act of 1956	1956-06-29	1	Spend- ing-dri- ven	Endoge- nous
Social Se- curity Amend- ments of 1956	1956-08-01	1	Spend- ing-dri- ven	Endoge- nous
Tax Rate Exten- sion Act of 1958	1958-06-30	1	Long-run	Exoge- nous

All Acts in us_shocks.csv

Act Name	Date Signed	Quarters	Motivation(s)	Exogeneity
Social Security Amendments of 1958	1958-08-28	2	Spending-driven, Deficit-driven	Endogenous, Exogenous
Federal-Aid Highway Act of 1959	1959-09-21	1	Deficit-driven	Exogenous
Social Security Amendments of 1961	1961-06-30	2	Spending-driven, Deficit-driven	Endogenous, Exogenous
Changes in Depreciation Guidelines and Revenue Act of 1962	1962-07-11	3	Long-run	Exogenous
Revenue Act of 1964	1964-02-26	2	Long-run	Exogenous
Excise Tax Reduction Act of 1965	1965-06-21	2	Long-run	Exogenous
Social Security Amendments	1965-07-30	2	Spending-driven	Endogenous

All Acts in us_shocks.csv

Act Name	Date Signed	Quarters	Motivation(s)	Exogeneity
ments of 1965				
Tax Ad- justment Act of 1966	1966-03-15	1	Long-run	Exoge- nous
Pub- lic Law 89-800 (Suspend- sion of Invest- ment Tax Credit)	1966-11-08	1	Counter- cyclical	Endoge- nous
Pub- lic Law 90-26 (Restora- tion of the In- vest- ment Tax Credit)	1967-06-13	1	Long-run	Exoge- nous
Social Se- curity Amend- ments of 1967	1968-01-02	3	Spend- ing-dri- ven, Deficit- driven	Endoge- nous, Ex- ogenous
Revenue and Ex- pendi- ture Con-	1968-06-28	2	Counter- cyclical	Endoge- nous

All Acts in us_shocks.csv

Act Name	Date Signed	Quarters	Motiva- tion(s)	Exogene- ity
trol Act of 1968				
Tax Re- form Act of 1969	1969-12-30	3	Counter- cyclical, Long-run	Endoge- nous, Ex- ogenous
1971 Changes to Social Security	1971-03-17	1	Spend- ing-dri- ven	Endoge- nous
Revenue Act of 1971	1971-12-10	1	Long-run	Exoge- nous
1972 Changes to Social Security	1972-07-01	2	Spend- ing-dri- ven, Deficit- driven	Endoge- nous, Ex- ogenous
1973 Changes to Social Security	1973-07-09	1	Spend- ing-dri- ven	Endoge- nous
Tax Re- duction Act of 1975	1975-03-29	2	Counter- cyclical	Endoge- nous
Tax Re- form Act of 1976	1976-10-04	2	Long-run	Exoge- nous
Tax Re- duction and Sim- plifica-	1977-05-23	1	Long-run	Exoge- nous

All Acts in us_shocks.csv

Act Name	Date Signed	Quarters	Motivation(s)	Exogeneity
tion Act of 1977				
Social Security Amendments of 1977	1977-12-20	4	Deficit-driven	Exogenous
Revenue Act of 1978	1978-11-06	1	Long-run	Exogenous
Crude Oil Windfall Profit Tax Act of 1980	1980-04-02	3	Long-run	Exogenous
Economic Recovery Tax Act of 1981	1981-08-13	4	Long-run	Exogenous
Tax Equity and Fiscal Responsibility Act of 1982	1982-09-03	1	Deficit-driven	Exogenous
Social Security Amendments of 1983	1983-04-20	5	Deficit-driven	Exogenous
Deficit Reduction	1984-07-18	1	Deficit-driven	Exogenous

All Acts in us_shocks.csv

Act Name	Date Signed	Quarters	Motivation(s)	Exogeneity
tion Act of 1984				
Tax Re-form Act of 1986	1986-10-22	4	Long-run	Exogenous
Omnibus Budget Reconciliation Act of 1987	1987-12-22	1	Deficit-driven	Exogenous
Omnibus Budget Reconciliation Act of 1990	1990-11-05	1	Deficit-driven	Exogenous
Omnibus Budget Reconciliation Act of 1993	1993-08-10	3	Deficit-driven	Exogenous
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	1997-08-05	3	Spending-driven, Deficit-driven	Endogenous, Exogenous
Economic Growth and Tax Relief Reconciliation	2001-06-07	2	Counter-cyclical, Long-run	Endogenous, Exogenous

All Acts in us_shocks.csv

Act Name	Date Signed	Quarters	Motivation(s)	Exogeneity
Reconciliation Act of 2001				
Job Creation and Worker Assistance Act of 2002	2002-03-09	1	Countercyclical	Endogenous
Jobs and Growth Tax Relief Reconciliation Act of 2003	2003-05-28	2	Long-run	Exogenous

3. Motivation Category Distribution

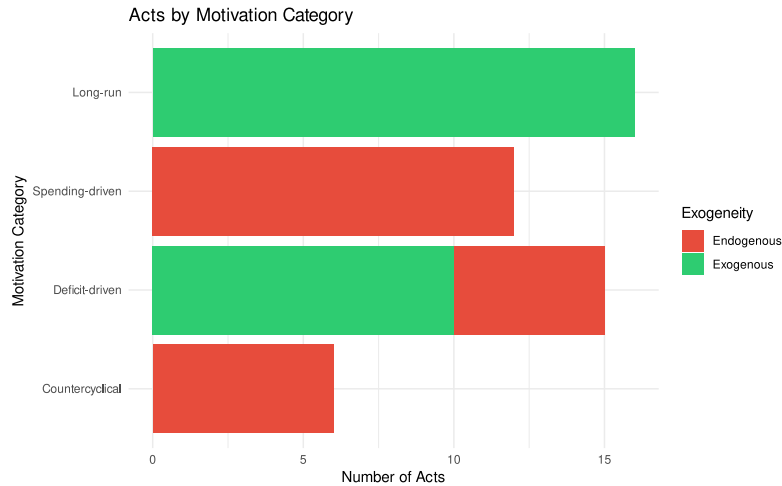
R&R classify each tax change along two dimensions: **exogeneity** (Exogenous vs. Endogenous) and **motivation** (Spending-driven, Countercyclical, Deficit-driven, Long-run). Only exogenous actions (deficit-driven and long-run) enter the instrumental variable for estimating fiscal multipliers. The distribution across acts determines the effective sample size for codebook evaluation.

Motivation Distribution Across Acts

Motivation	Exogeneity	Number of Acts
Countercyclical	Endogenous	6
Deficit-driven	Endogenous	5
Spending-driven	Endogenous	12
Deficit-driven	Exogenous	10
Long-run	Exogenous	16

The split is **26 exogenous** vs. **23 endogenous** acts. Among exogenous acts, long-run motivations dominate (16 acts), followed by

deficit-driven (10 acts). No exogenous acts are spending-driven or countercyclical, which is consistent with R&R's definitions: those motivations inherently respond to the business cycle.



4. Row-Level Motivation Breakdown

Some acts contain rows with different motivation categories. This happens when a single piece of legislation has provisions with distinct economic rationales: for example, a Social Security amendment may pair an immediate spending-driven tax increase with a phased deficit-driven increase years later, or EGTRRA 2001 combines a countercyclical rebate with long-run rate reductions.

Acts with Mixed Motivations

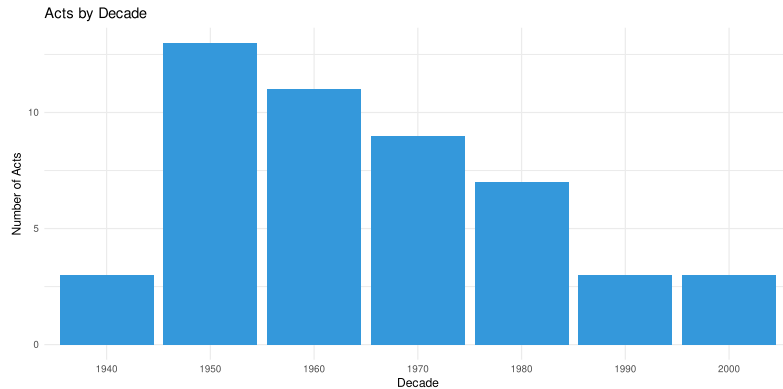
Act Name	Distinct Categories	Exogeneity
1972 Changes to Social Security	Spending-driven + Deficit-driven	2
Economic Growth and Tax Relief Reconciliation Act of 2001	Countercyclical + Long-run	2

Acts with Mixed Motivations

Act Name	Distinct Categories	Distinct Exogeneity
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	2 Spending-driven + Deficit-driven	2
Social Security Amendments of 1950	2 Spending-driven + Deficit-driven	2
Social Security Amendments of 1958	2 Spending-driven + Deficit-driven	2
Social Security Amendments of 1961	2 Spending-driven + Deficit-driven	2
Social Security Amendments of 1967	2 Spending-driven + Deficit-driven	2
Tax Reform Act of 1969	2 Counter-cyclical + Long-run	2
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	2 Spending-driven + Deficit-driven	2

9 of 49 acts (18%) have mixed motivations. All 9 also have mixed exogeneity, meaning they contain both exogenous and endogenous components. These acts are the hardest cases for codebook C2 (motivation classification) because an LLM must distinguish which provisions fall under which motivation. They will serve as important test cases during S1 behavioral testing and S2 evaluation.

5. Temporal Coverage



Coverage peaks in the 1950s (13 acts) and 1960s (11 acts), reflecting frequent postwar tax legislation. Coverage thins in the 1990s and 2000s (3 acts each), though the later acts tend to be larger in dollar magnitude. The 1940s have only 3 acts because the sample begins in 1945. This uneven temporal distribution means LOOCV folds will have varying amounts of training data; codebook S2 evaluation should track performance by decade to detect any systematic bias.

6. Labels-Shocks Cross-Reference

Act Coverage Across Datasets

Category	Count
In both datasets	40
In us_shocks only	9
In us_labels only	0

Acts in us_shocks but NOT in us_labels:

- Social Security Amendments of 1947
- Revenue Act of 1951
- Social Security Amendments of 1956
- Social Security Amendments of 1958
- Social Security Amendments of 1961

- Social Security Amendments of 1967
- 1971 Changes to Social Security
- 1972 Changes to Social Security
- 1973 Changes to Social Security

40 of 49 acts appear in both datasets. The 9 acts with shock data but no extracted labels are mostly Social Security amendments and highway acts. Their companion paper narratives are shorter and contain fewer direct quotations, so the quote-extraction regex found no passages meeting the minimum length threshold. This is not a data error: the shock data (quarters, amounts, categories) is complete for all 49 acts regardless of whether quotation evidence was extracted. For codebook C2 evaluation, the full narrative text (stored in the Reasoning column of `us_shocks`) provides the classification evidence even for acts without separately extracted quotes.

7. Label Consistency Check

For acts present in both datasets, we verify that the motivation category and exogeneity agree. Because `us_labels` carries only the **primary** classification per act (from the first standard entry), while `us_shocks` may contain multiple categories for mixed-motivation acts, we check whether the label classifications are a subset of the shock classifications.

All label classifications are consistent with their corresponding shock classifications.

8. Labels Evidence Summary

The evidence passages in `us_labels` are direct quotations from presidential speeches, budget messages, committee reports, and economic reports that R&R cite to justify their motivation classifications. These passages will serve as few-shot examples for codebook C2.

Evidence Passages per Act (`us_labels`)

Act Name	Passages
Tax Reform Act of 1969	26
Jobs and Growth Tax Relief Reconciliation Act of 2003	21
Economic Recovery Tax Act of 1981	19

Evidence Passages per Act (us_labels)

Act Name	Passages
Revenue Act of 1964	16
Revenue and Expenditure Control Act of 1968	15
Deficit Reduction Act of 1984	14
Job Creation and Worker Assistance Act of 2002	14
Revenue Act of 1945	14
Tax Reduction Act of 1975	14
Public Law 89-800 (Suspension of Investment Tax Credit)	13
Revenue Act of 1971	13
Omnibus Budget Reconciliation Act of 1987	12
Public Law 90-26 (Restoration of the Investment Tax Credit)	11
Tax Equity and Fiscal Responsibility Act of 1982	11
Revenue Act of 1948	10
Crude Oil Windfall Profit Tax Act of 1980	9
Tax Reform Act of 1976	9
Excise Tax Reduction Act of 1954	8
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	8
Internal Revenue Code of 1954	8
Omnibus Budget Reconciliation Act of 1993	8
Revenue Act of 1950	8
Tax Rate Extension Act of 1958	8
Tax Reform Act of 1986	8
Changes in Depreciation Guidelines and Revenue Act of 1962	7
Excise Tax Reduction Act of 1965	7
Federal-Aid Highway Act of 1956	7
Tax Reduction and Simplification Act of 1977	7

Evidence Passages per Act (us_labels)

Act Name	Passages
Economic Growth and Tax Relief Reconciliation Act of 2001	6
Omnibus Budget Reconciliation Act of 1990	6
Revenue Act of 1978	6
Social Security Amendments of 1965	6
Excess Profits Tax Act of 1950	5
Federal-Aid Highway Act of 1959	4
Social Security Amendments of 1977	4
Tax Adjustment Act of 1966	4
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	4
Social Security Amendments of 1950	2
Social Security Amendments of 1954	2
Social Security Amendments of 1983	2

Evidence Passage Statistics

Min	Median	Mean	Max	Total
2	8	9.4	26	376

The median act has **8 passages** (mean 9.3), with a range from 1 to 26. The Tax Reform Act of 1969 has the most passages (26), reflecting its complex legislative history and mixed motivations. Acts with fewer passages tend to have straightforward, single-motivation narratives.

9. Data Quality Flags

No data quality issues found. All fields are populated and consistent.

Summary and Assessment

Training Data Audit Summary

Metric	Value
Unique acts in us_shocks	49
Unique acts in us_labels	40
Acts in both datasets	40
R&R companion paper act headings	49
Coverage	100%
Exogenous acts (primary)	26
Endogenous acts (primary)	23
Mixed-motivation acts	9
Total evidence passages	376
Data quality issues	0

The data is ready for codebook development. All 49 R&R act headings are present with complete shock data (quarters, amounts, categories, present values). The key properties that support Phase 0 evaluation are:

- **Full R&R coverage.** 100% of companion paper act headings are captured, providing the complete benchmark against which codebooks C1–C4 will be evaluated.
- **Clean, consistent categories.** Four motivation types and two exogeneity types are consistently applied across all 90 rows with no missing values.
- **Balanced exogeneity split.** The 26/23 exogenous/endogenous split provides adequate representation of both classes for LOOCV evaluation. C2 codebook precision targets (Exogenous Precision $\geq 85\%$) are evaluable against 26 exogenous acts.
- **Mixed-motivation complexity.** 9 acts (18%) have mixed motivations, providing hard test cases for C2 behavioral testing (S1) and error analysis (S3).
- **Rich evidence base.** 372 extracted quotations across 40 acts supply the few-shot examples and ground-truth reasoning for codebook development.

i Note

The 9 acts without extracted label passages still have complete narrative text in the Reasoning column of `us_shocks`. Codebook C2 evaluation uses the full act narrative, not the separately extracted quotes. The `us_labels` dataset primarily supports few-shot example construction and motivation-evidence traceability.